Solution Requirements

Date	30 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	4 Marks

Project Overview

Smart Lender is a machine learning-powered web application that predicts the creditworthiness of loan applicants, enabling banks and financial institutions to make faster, data-driven loan approval decisions. The platform evaluates applicant data using classification algorithms (Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost) to determine the likelihood of loan repayment or default. The best-performing model, XGBoost, is integrated into a Flask web application for real-time prediction, with deployment planned on IBM Cloud.

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority
1	**Authentication**	Users (credit officers, financial analysts, administrators) log in to the Smart Lender web application using a registered email address and password. Passwords are securely hashed before storage, and session-based authentication is used to keep users logged in throughout their evaluation workflow.	High
2	**Authorization levels**	Three roles are supported: Admin (manages users, retrains models, views all data), Credit Officer / Analyst (submits applicant details, views predictions and reports for their own submissions), and Applicant/Guest (can submit a single application form and view only their own result). Access to batch evaluation and dashboard analytics is restricted to Credit Officer and Admin roles.	High
3	**External interfaces**	The system connects to a trained XGBoost model file (serialized with Pickle/Joblib) loaded by the Flask backend for real-time inference. It is designed for deployment on IBM Cloud, and may integrate with an external relational database (e.g., IBM Db2 or MySQL) to store applicant records and prediction history.	Medium
4	**Transactions processing**	Each loan evaluation request is processed as a single transaction: applicant inputs (gender, marital status, education, employment status, income, loan amount, loan term, credit history, property area) are validated, passed to the trained model, and the prediction (Approved/Rejected) along with a confidence score is returned and logged in real time.	High
5	**Reporting**	The system generates individual prediction reports for each	Medium

S. No	Requirement Category	Requirement Description	Priority
		applicant, a downloadable summary report for batch-evaluated applicants, and an analyst dashboard showing approval/rejection trends, model accuracy metrics, and high-risk applicant counts over a selected time period.	
6	**Business rules, etc.**	Applications with no credit history or irregular self-employment income are automatically flagged as high-risk and routed for manual document verification rather than auto-approval. Applicants with a strong credit history and stable salaried income that meet the model's high-confidence threshold may be fast-tracked without manual review.	High
7	**Compliance to laws or regulations**	The system adheres to applicable data privacy and financial-sector regulations (e.g., RBI guidelines on digital lending and applicant data protection, and general data protection principles similar to GDPR) for the secure handling, storage, and retention of applicant financial and personal data.	High
8	**Other**	Provision for periodic retraining of the model with new applicant data, an audit trail of all predictions, and an exportable CSV/PDF of batch evaluation results for the financial analyst's records.	Low

Step 2: Non-Functional Requirements (NFR)

S. No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	**Performance & Speed**	Real-time prediction must return a result quickly after an applicant's details are submitted, supporting fast-track decisions during high-volume periods.	Prediction response time < 2 seconds per applicant
2	**Scalability**	The system must support growth in the number of applicants and concurrent credit officers, including bulk/batch evaluation during peak loan-application periods.	Supports up to 1,000 concurrent users and batch evaluation of 5,000+ applicant records
3	**Security & Data Privacy**	Applicant personal and financial data (income, credit history, etc.) must be encrypted in transit and at rest, with role-based access control to prevent unauthorized viewing of sensitive applicant records.	TLS 1.2+ in transit; AES-256 encryption at rest; role-based access control enforced
4	**Reliability & Availability**	The platform must remain available for credit officers and analysts during business hours, with graceful handling of model-loading or prediction failures.	High uptime per month; automatic fallback/error message on prediction failure
5	**Usability & Accessibility**	The applicant submission form and analyst dashboard must be simple and intuitive for non-technical banking staff to use, with clear guidance on required fields.	Complies with WCAG 2.1 AA standards; new users complete a submission in under 3 minutes without training
6	**Other**	The codebase and model pipeline (NumPy, Pandas, Scikit-learn, XGBoost, Flask) must be maintainable and portable across environments, and deployable consistently on IBM Cloud.	Maintainability: modular Flask app structure; Portability: deployable on IBM Cloud and local environments with no code changes